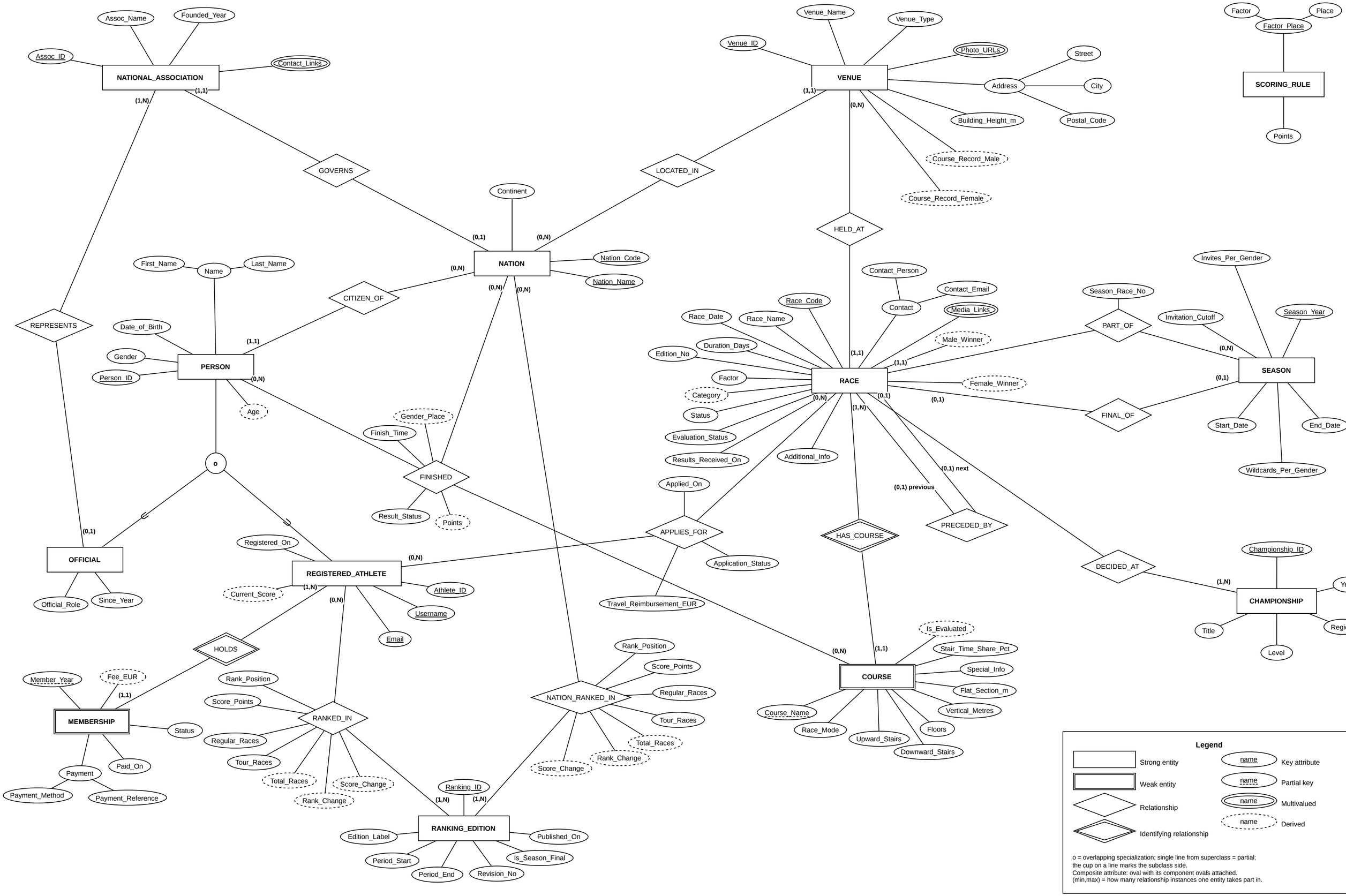
EER diagram: Towerrunning World Database (Team 63 requirements), modelled by Team 59

Elmasri & Navathe notation. Every relationship edge carries (min,max) beside the entity it constrains.



Phase 2 — EER Model of Team 63's Tower Running Requirements

Team 59 (Aggregays) · Assigned document: Team 63 (Carromboard), *Project Phase 1: Data Requirements — Tower Running* (domain https://www.towerrunning.com)

Our EER model keeps all 13 entity types and 15 relationship types of Team 63's Towerrunning World Database (TWDB), including the ternary FINISHED. The diagram is on the first page; the sections below give our critical evaluation, the changes we made, our notational deviations and the design quality analysis. Section and rule numbers in brackets, such as [2.4 rule 4], point to Team 63's document.

1. EER diagram

The diagram on page 1 has 13 entity types, 15 relationship types and 110 attributes. Each (min,max) pair sits beside the entity it constrains and counts how many relationship instances one entity of that type takes part in. The table lists every relationship with the line in Team 63's document that fixes its numbers.

#	Relationship	Participants with (min,max)	Taken from
R1	CITIZEN_OF	PERSON (1,1), NATION (0,N)	2.3 R1: N:1, PERSON total
R2	LOCATED_IN	VENUE (1,1), NATION (0,N)	2.3 R2: N:1, VENUE total
R3	HELD_AT	RACE (1,1), VENUE (0,N)	2.3 R3: one venue per race
R4	HAS_COURSE (identifying)	RACE (1,N), COURSE (1,1)	2.3 R4: both total
R5	HOLDS (identifying)	REGISTERED_ATHLETE (1,N), MEMBERSHIP (1,1)	2.3 R5: both total
R6	PART_OF	RACE (1,1), SEASON (0,N)	2.3 R6: RACE total
R7	FINAL_OF	SEASON (0,1), RACE (0,1)	2.3 R7: 1:1, both partial
R8	DECIDED_AT	CHAMPIONSHIP (1,N), RACE (0,1)	2.3 R8 and the "one championship per race" assumption in 1.1
R9	GOVERNS	NATIONAL_ASSOCIATION (1,1), NATION (0,1)	2.3 R9: 1:1, association total
R10	REPRESENTS	OFFICIAL (0,1), NATIONAL_ASSOCIATION (1,N)	2.3 R10: N:1, association total
R11	RANKED_IN	REGISTERED_ATHLETE (0,N), RANKING_EDITION (1,N)	2.3 R11: M:N, edition total
R12	NATION_RANKED_IN	NATION (0,N), RANKING_EDITION (1,N)	2.3 R12: M:N, edition total
R13	APPLIES_FOR	REGISTERED_ATHLETE (0,N), RACE (0,N)	2.3 R13: M:N, both partial
R14	PRECEDED_BY (recursive)	RACE as next (0,1), RACE as previous (0,1)	2.3 R14: 1:1, both partial
R15	FINISHED (ternary)	PERSON (0,N), COURSE (0,N), NATION (0,N)	2.3 R15: all partial; see finding C2

The specialization of PERSON into OFFICIAL and REGISTERED_ATHLETE is overlapping (o) and partial, as 2.2 states. SCORING_RULE stands alone with no relationship (finding C1).

2. Critical evaluation of Team 63's document

The document is well grounded and mostly modelable as written. We found eight completeness gaps, seven correctness problems (four of them contradicted by the live site) and six places where the prose did not translate cleanly into EER. Each fix below says whether it changes the diagram.

2.1 Completeness

C1. SCORING_RULE has no relationship [2.1, 2.3]. None of R1–R15 involves it; it appears only inside the Points derivation [2.4]. Points uses (race Factor, Gender_Place), and Gender_Place is itself derived, so no stored foreign key can point at a rule.

- Fix: kept as a stand-alone lookup entity. In Phase 3, a finisher's Points come from the SCORING_RULE row whose Factor equals the race's Factor and whose Place equals the finisher's Gender_Place. Not a diagram change.

C2. The ternary FINISHED has no cardinality [2.3 R15]. The table gives only "all partial". The text below it adds that each (PERSON, COURSE) pair has exactly one NATION, which (min,max) cannot express on a ternary.

- Fix: (0,N) on all three participants; the (PERSON, COURSE) → NATION rule is kept as a constraint, matching the key (Person_ID, Race_Code, Course_Name) they state.

C3. Many attributes have no domain [2.4]. Missing: Official_Role, Since_Year, Venue_Type, Building_Height_m, Founded_Year, Title and Year of CHAMPIONSHIP, Edition_No, Flat_Section_m, Applied_On, Published_On and Travel_Reimbursement_EUR.

- Fix (domains only): Official_Role ∈ {President, Sports Director, Presidium member, National representative}; Venue_Type ∈ {Skyscraper, Tower, Outdoor staircase}, the three kinds named in their opening paragraph; years are 4-digit, and Since_Year and Founded_Year are not in the future; heights and metres > 0; Edition_No is a positive integer counting editions, not a year; Flat_Section_m is the total length of flat sections in metres, ≥ 0, with 0 for a course that has none; Applied_On and Published_On are dates; Title is non-empty text; Travel_Reimbursement_EUR ∈ {0, 600}, from the Tour Final row in 1.1, and is 600 only when Application_Status is Invited, Wild card or Accepted. These are our proposed domains, not rules Team 63 states.

C4. RANKING_EDITION has no natural key [2.1]. Only Ranking_ID is a key, so two "July 2026" editions with the same Revision_No could both be stored.

- Fix: (Period_Start, Period_End, Revision_No) is unique, since two editions covering the same window at the same revision would be duplicates, and Period_Start ≤ Period_End. We do not make (Period_End, Revision_No) unique on its own: an intermediate edition and a regular one could share a cut-off date. Constraints, not drawn.

C5. Organisers write data but are never stored [1.2]. Event organisers may edit "own races", but RACE holds only a free-text Contact. Nothing links a login to the races it may change, so this access rule cannot be enforced in Phase 3.

- Not fixed, to keep fidelity. Flagged for Phase 3.

C6. Date_of_Birth is not marked optional [2.1 PERSON]. The Results Form collects only name, gender and nationality [1.1], so most unregistered finishers have no birth date, and Age cannot be derived for them.

- Fix: Date_of_Birth is optional on PERSON and required for REGISTERED_ATHLETE, which I1 already enforces. Not a diagram change.

C7. Nothing ties a Final to its own season [2.3 R6, R7]. A race could be FINAL_OF season 2026 while PART_OF season 2025.

- Fix: added the constraint that the FINAL_OF race is PART_OF the same season.

C8. Ranking rows lose the eight counted results [1.1 Ranking row vs. 2.3 R11]. Team 63's own table says each ranking row shows the athlete's best 8 results and the points won at every race in the window, but RANKED_IN stores only totals. Because editions are frozen and results get corrected, a past edition's counted races cannot be rebuilt later.

- Not fixed, to keep fidelity. Flagged for Phase 3, where a weak entity for counted results would close the gap.

2.2 Correctness

K1. Ranking editions are not always monthly [2.1 RANKING_EDITION, 2.4 Ranking_Month ∈ 1–12]. The Ranking page (https://www.towerrunning.com/ranking/) lists "TWA Ranking AprilMay10 2026", cut off on 10 May, and "TWA Ranking Intermediate Jan 21 – Mar 22", which spans 15 months. Neither fits a single Ranking_Month and Ranking_Year.

- Fix: replaced Ranking_Month and Ranking_Year with Period_Start, Period_End and Edition_Label. This is a diagram change.

K2. A 2020 ranking does exist [1.1, Statistics row]. The document says there is no 2020 season ranking because of the pandemic, but the same Ranking page lists "TWA Ranking 2020 Final".

- No model change. The scope statement is wrong, not the design.

K3. The Athlete-ID cannot be both a primary key and issued later [2.2, 2.6 vs. 3.3 I1]. Athlete_ID is the primary key of REGISTERED_ATHLETE, so it must exist on insert. Yet I1 inserts bank-transfer registrants as Pending and says the Athlete-ID is "not released" until verified, and R5's note says an Athlete-ID exists "only after a paid membership".

- Fix: we read "released" as "sent to the athlete". The ID is generated on insert, and HOLDS stays (1,N) because the Pending MEMBERSHIP row exists from the start. R5's note is corrected to "after a membership is created".

K4. Stair_Time_Share_Pct cannot be measured [2.4 rule 1 vs. 1.1 scope]. Rule 1 requires it to exceed 50, but the scope says split times are never published and the Results Form carries only a total time.

- Fix: kept the attribute, but defined as the share the organiser declares when the course is registered.

K5. Regular-race factor is known before the race [2.4 rule 4]. Rule 4 keeps a regular race's Factor NULL until evaluation. The Ranking page, however, offers a separate "Results Form Factor 40" for these races, so organisers know the factor in advance.

- No model change: a NULL Factor still means "not yet assigned". We note that 40 could be set at creation instead.

K6. Results can be merged into the wrong person [3.3 I2]. I2 links a results line to an existing PERSON when full name, gender and nation match. That treats (name, gender, nation) as a key, which it is not: two athletes from one country can share a name.

- Fix: a match is only a suggestion the administrator confirms; nothing is linked automatically. Not a diagram change.

K7. Tour Final invitation rules change every year [2.4 rule 11]. Rule 11 fixes 24 invitations from the October ranking, which is the 2026 rule. The 2025 Final gave 20 invitations, the top 8 per gender of the September 2025 ranking plus 2 wild cards per gender (https://www.towerrunning.com/2025/10/09/towerrunning-tour-final-2025-shanghai-tower-november-22/). Rule 11 cannot describe that season: its invitees came from a different ranking and a different cut.

- Fix: added Invites_Per_Gender and Wildcards_Per_Gender (non-negative integers) and Invitation_Cutoff (a date within the season) to SEASON, and rule 11 reads them. This is a diagram change.

2.3 Modelability

M1. Derived attributes depend on other entities. Points needs SCORING_RULE, the race's Factor and Gender_Place; Current_Score needs a 12-month window. A dashed oval shows only that a value is derived; the rule itself stays in the text.

M2. The ternary hides a functional dependency. (min,max) on FINISHED bounds each participant separately, so "one nation per person per course" cannot be drawn. We kept the ternary, because the document chose it and it is the required n ≥ 3 relationship; their own bonus (2.6) shows the RESULT weak-entity alternative.

M3. The subclass has its own keys. REGISTERED_ATHLETE carries three keys (Athlete_ID, Username, Email) besides the inherited Person_ID. EER underlines all keys equally, so the choice of Athlete_ID as primary key [2.6] is not visible on the diagram.

M4. "Same attributes as RANKED_IN" [2.3 R12]. NATION_RANKED_IN's attributes are given only by reference. We drew all seven on both relationships, which doubles 14 ovals in the busiest corner.

M5. The composite key has no name [2.1 SCORING_RULE {Factor, Place}]. Elmasri & Navathe draw a key made of several attributes as one composite attribute with the whole group underlined, as with CAR's Registration (State, Number). Underlining Factor and Place separately would instead say each is a key on its own, which is false. The group needs a name, so we called it Factor_Place.

M6. Rules that no diagram can show. Rules 5, 6, 8, 10 and 11 in 2.4, and the FINAL_OF/PART_OF rule from C7, are cross-entity constraints. They are listed here for Phase 3 triggers, not drawn.

3. Changes beyond the evaluation

None. The diagram departs from Team 63's document in only three places, all listed above. K1 replaces Ranking_Month and Ranking_Year with Period_Start, Period_End and Edition_Label. K7 adds Invites_Per_Gender, Wildcards_Per_Gender and Invitation_Cutoff to SEASON. M5 names the composite key Factor_Place. Every other fix is a domain, a constraint or a clarified meaning, so the diagram's entities, relationships and attributes otherwise match the document one for one.

4. Notational deviations

Deviation	Reason
Total participation is shown by (min,max) only, with single lines rather than double lines	The brief asks for (min,max) on every edge; Elmasri & Navathe treat (min,max) as an alternative to the double-line notation, and mixing both would say the same thing twice
Role names of the recursive PRECEDED_BY (next, previous) are written next to their (min,max) labels	Keeps each line's role and constraint in one place
The composite key of SCORING_RULE is named Factor_Place	A composite attribute needs a name to underline; Team 63's document gives none
Attribute and type names use underscores instead of spaces	They follow Team 63's spelling exactly, which eases the Phase 3 mapping
The diagram page is 20 × 14 in rather than A4	110 attributes do not fit legibly on A4; the page is vector and can be zoomed

5. Design quality analysis

Our model does contain redundancy and all three kinds of anomaly. Some are deliberate choices inherited from Team 63; the rest come from storing each race edition as a separate RACE with no entity for the event itself.

5.1 Redundancy

1. **The Final's season is stored twice.** A Tour Final is linked to its season through FINAL_OF and again through PART_OF. Nothing but the constraint from C7 keeps the two equal.
2. **A finisher's nation overlaps with citizenship.** FINISHED records the race-day nation and CITIZEN_OF the current one. These are different facts, so this is intentional overlap rather than true duplication, but for most athletes the two values match and nothing keeps them consistent.
3. **Ranking rows repeat what results already imply.** Score_Points and Rank_Position on RANKED_IN and NATION_RANKED_IN can be recomputed from FINISHED, COURSE, RACE and

SCORING_RULE. They are frozen snapshots, because corrections and the moving 12-month window would otherwise change past editions.

4. **The season-final flag partly repeats the dates.** Is_Season_Final is stored, not derived, because the TWA designates the year-end edition (the "Final 2025" file) and a corrected reissue can cover the same dates. It still overlaps with what Period_End and the SEASON dates imply, so the flag and the dates can disagree.

5.2 Update anomaly

The clearest case is renaming an event. Each RACE row is one edition, so "Empire State Building Run Up" is stored once per year. Renaming the event means updating Race_Name on every edition, reached by walking the PRECEDED_BY chain; missing one leaves two names for one event.

A second case is postponing a Final into the next season (Team 63's U3). PART_OF and FINAL_OF must both change, and Season_Race_No must be renumbered within the new season.

5.3 Insertion anomaly

A newly recognised national association cannot be stored until it has a representative. NATIONAL_ASSOCIATION has (1,N) in REPRESENTS, so the TWA cannot record, say, a new Indian association before naming its OFFICIAL.

Similarly, CHAMPIONSHIP has (1,N) in DECIDED_AT. A 2027 World Championship that has been announced cannot be inserted until its deciding race exists with a Race_Code.

5.4 Deletion anomaly

Deleting a cancelled race (Team 63's D1) also deletes its COURSE rows, because COURSE is weak. This loses the route's Upward_Stairs, Floors and Vertical_Metres. If a course is meant to carry over to later editions, as when a race climbs the same building each year, those facts had nothing to do with the cancellation and are lost with it.

In the same way, removing the only OFFICIAL who represents an association breaks that association's (1,N) participation. The delete must either be refused or take the association's record with it.

6. Use of generative AI

We used Claude (Anthropic) to summarise Team 63's document, suggest a direction, explain concepts, and help us debug and understand how to approach the work.